

Avinash Nallala

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EDUCATION

Master of Science in Data Analytics**GPA:3.78****Dec 2025**

Clark University; Worcester, MA

Bachelor of Technology in Electronics & Communication Engineering**GPA:3.2****May 2022**

Mahaveer Science & Technology; Hyderabad, TS, INDIA

PROFESSIONAL SUMMARY

Machine Learning Engineer and Data Analyst with hands-on experience in building end-to-end data-driven solutions, including data preprocessing, exploratory data analysis (EDA), model development, and deployment. Proficient in Python and SQL with expertise in supervised and unsupervised learning techniques such as regression, classification, clustering, and ensemble methods including Random Forest and XGBoost. Experienced in feature engineering, model evaluation (R^2 , RMSE, accuracy, precision, recall), and hyperparameter tuning using GridSearchCV and RandomizedSearchCV. Skilled in developing REST APIs using Flask and deploying machine learning models on cloud platforms such as AWS and Render. Strong background in data visualization using Tableau and Power BI to deliver actionable insights.

RELEVANT SKILLS

- **Programming & Querying:** Python (NumPy, Pandas), SQL (Joins, Aggregations, Window Functions, CTEs)
- **Machine Learning & Modeling:** Supervised Learning (Linear Regression, Logistic Regression, Decision Trees, Random Forest, KNN, Naive Bayes), Unsupervised Learning (K-Means Clustering, DBSCAN, Hierarchical Clustering, PCA), Feature Engineering, Feature Selection, Model Evaluation (R^2 , RMSE, Accuracy, Precision, Recall), Hyperparameter Tuning (GridSearchCV, RandomizedSearchCV), Scikit-learn Pipelines
- **Statistical Analysis:** Hypothesis Testing, Probability Distributions, Correlation Analysis, A/B Testing Basics
- **Data Analysis & Visualization:** Exploratory Data Analysis (EDA), Data Cleaning, Data Transformation, Matplotlib, Seaborn, Tableau, Power BI (Dashboards, KPIs, Filters, Slicers)
- **Backend & API Development:** Flask, Django, REST API Development, JSON Handling, Request/Response Lifecycle, API Testing (Postman, Python Requests)
- **Databases & Data Management:** MySQL (CRUD Operations, Indexing, Query Optimization, Schema Design)
- **Data Engineering:** ETL Pipelines, Data Transformation, Handling Missing Data
- **Core Computer Science:** Data Structures & Algorithms (Arrays, Linked Lists, Trees, Hashing), System Design (Client-Server Architecture, REST Architecture)
- **Deployment & MLOps:** Model Deployment using Flask APIs, End-to-End ML Deployment Workflow, Git, GitHub, Render
- **Tools & Environment:** VS Code, Jupyter Notebook, Postman

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Machine Learning – Coursera
- Microsoft Power BI Data Analyst (PL-300)
- Google Data Analytics Professional Certificate

EXPERIENCE

Machine Learning Engineer | Devscape IT Solutions Virginia USA | May 2024 - Feb 2026

House Price Prediction System

- Developed an end-to-end machine learning model to predict house prices using regression algorithms and XGBoost
- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA) using Pandas and NumPy
- Evaluated model performance using R^2 score, RMSE, and cross-validation techniques
- Built a REST API using Flask to serve real-time predictions
- Deployed the application on a cloud platform (Render/AWS) for public access
- Created interactive dashboards using Power BI/Tableau to visualize pricing trends and insights

Customer Segmentation

Customer Segmentation using Unsupervised Learning

- Implemented K-Means clustering and PCA for customer segmentation and pattern recognition
- Conducted data cleaning, scaling, and exploratory data analysis (EDA)
- Visualized clusters and relationships using Matplotlib and Seaborn
- Built interactive dashboards using Power BI/Tableau to analyze customer behavior and trends
- Derived actionable insights to support business decision-making

Machine Learning Engineer | ZipNom Technologies Hyderabad India | Jan 2021 – Sep 2023

Credit Risk & Loan Approval Prediction System

- Developed an end-to-end machine learning model to predict loan approval using classification algorithms including Logistic Regression, Random Forest, and XGBoost
- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA) to identify key risk factors
- Evaluated model performance using accuracy, precision, recall, F1-score, and confusion matrix
- Built RESTful API using Flask to serve real-time predictions
- Deployed the application on cloud platform (Render/AWS) for scalability and accessibility
- Designed interactive dashboards using Power BI/Tableau to analyze credit risk and approval trends

E-Commerce Recommendation System

- Developed a recommendation system using collaborative and content-based filtering techniques to suggest products based on user preferences
- Performed data preprocessing, feature engineering, and similarity analysis using cosine similarity
- Built a Flask-based REST API to provide real-time product recommendations
- Deployed the application on cloud platform (Render/AWS) for scalability
- Created interactive dashboards using Power BI/Tableau to analyze user behavior and product trends